

COURSEMATERIAL

SUBJECT	ENGINEERINGPHYSICS(PH23ABS101)
UNIT	IV
COURSE	B.TECH
DEPARTMENT	S&H
SEMESTER	II
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TABLEOFCONTENTS–UNITIV

S.NO	CONTENTS	PAGENO.
1	COURSEOBJECTIVES	1
2	PREREQUISITES	1
3	SYLLABUS	1
4	COURSEOUTCOMES	1
5	CO-PO/PSOMAPPING	2
6	LESSONPLAN	2
7	LECTURENOTES	3
	7.1 INTRODUCTION	3
	7.2 DUALNATUREOFMATTER	7
	7.3 HEISENBERGUNCERTAINTYPRINCIPLE	8
	7.4 PHYSICALSIGNIFICANCEOFWAVEFUNCTION	10
	7.5 SCHRODINGER'SWAVEEQUATION	14
	7.6 PARTICLEINONEDIMENSIONALBOXORWELL	19
	7.7 INRODUCTIONTOFREEELECTRONTHEORY	19
	7.8 MERITSOFCCLASSICALFREEELECTRONTHEORY	20
	7.9 DEMERITSOFCCLASSICALFREEELECTRONTHEORY	20
	7.10 QUANTUMFREEELECTRONTHEORY	25
	7.11 FERMIDIRACDISTRIBUTION	28
8	PRACTICEQUIZ	38
9	ASSIGNMENTS	39

10	PART A QUESTIONS & ANSWERS (2 MARKS QUESTIONS)	39
11	PART B QUESTIONS	42
12	PRESCRIBED TEXT BOOKS & REFERENCE BOOKS	43



1. Course Objectives

Bridge the gap between the Physics in school at 10+2 level and UG level engineering courses by identifying the importance of the optical phenomenon like interference, diffraction etc, enlightening the periodic arrangement of atoms in crystalline solids and concepts of quantum mechanics, introduce novel concepts of dielectric and magnetic materials, physics of semiconductors.

Prerequisites

Students should have knowledge on the basics of

1. Quantum Mechanics
2. Free electron theory
3. Band Theory of solids

2. Syllabus

UNIT IV

Quantum Mechanics-Dual nature of matter–

Heisenberg uncertainty principle (Qualitative) – Schrodinger's time independent and dependent wave equation – Significance of wave function – Particle in a one-dimensional infinite potential well.

Free Electron Theory- Classical free electron theory (Merits and demerits only) – Quantum free electron theory – Equation for electrical conductivity based on quantum free electron theory–Fermi-Dirac distribution.

3. Course outcomes

- CO 1: Understand the intensity variation of light due to interference, diffraction and polarization.
- CO 2: Apply the basic concepts of crystal structures and X-ray diffraction to study the behavior of materials for engineering applications.
- CO 3: Summarize the fundamental properties of dielectric and magnetic materials for engineering applications.
- CO 4: Analyze the properties of quantum particles to interpret the energy band formation and classification of solids
- CO 5: Assess the current flow mechanism to understand the transport phenomenon of charge carriers in semiconductors.

4. Co-PO/PSOMapping

APPLIED PHYSICS	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	P10	PO11	PO12	PSO1	PSO2
CO1	3	3												
CO2	3	3												
CO3	3	3												
CO4	3	3												
CO5	3	3												

5. LessonPlan

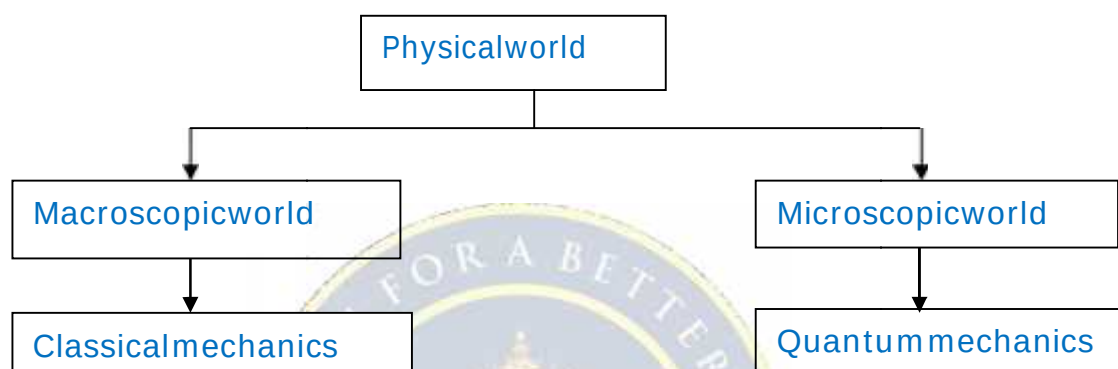
Lecture No.	Weeks	Topics to be covered	References
1	1	Quantum Mechanics: Matter waves, Dual nature of matter	T1, R1
2		Heisenberg uncertainty	
3		principle (Qualitative) Schrodinger's time dependent	T1,
4		wave	R1 T1,
5		equations Schrodinger's time independent wave equations	R1
6	2	Physical significance of wave function, Particle in one dimensional infinite potential well.	T1, R1
7		Free electron theory: Classical free electron theory (Merits and demerits)	T1,
8		Quantum free electron theory	R1 T1,
9		Equation for electrical conductivity based on quantum free electron theory	R1 T1,
10	3		R1
11		Fermi-Dirac distribution, Source of electrical resistance	T1,
12		Classification of solids into conductors, semiconductors and insulators.	R1 T1,
13			R1 T1,

R1

14	4		T1, R1
15		A review on quantum mechanics and free electron theory	T1, R1
16		Problems-1	T1, R1

7. Lecture Notes

7.1. Introduction



In the late seventeenth century, Isaac Newton discovered classical mechanics, the law of motion of macroscopic objects.

In the early twentieth century, physicists found that classical mechanics does not correctly describe the behavior of very small particles such as the electrons nuclei (protons & neutrons etc.) and molecules. The behavior of such particles is described by a set of laws called Quantum mechanics.

I.e. Classical mechanics applies only to macroscopic particles. For microscopic “particles” we require a new form of mechanics, called Quantum mechanics.

Classical mechanics:

- Classical mechanics deals with the motion and behavior of macroscopic particles.
Ex: Motion of planets.
- Classical mechanics is expressed in the language of fun-probabilities, certainties.

Quantum mechanics:

- Quantum mechanics deals with the motion and behavior of microscopic particles

Ex: Motion of Electrons

- Quantum mechanics is expressed in the language of probabilities, not certain.

7.2. Dual Nature of Matter

7.2.1 de-Broglie hypothesis

According to de-Broglie hypothesis, every moving particle has always associated with a wave. The particle is controlled by this wave in every aspect. The wave associated with moving particle of matter (like electrons, protons etc) are known as matter wave or pilot Broglie waves.



Figure.1 Motion of particle (Electron)

If a particle of mass m moving with velocity v , then the wavelength of matter wave is given by

$$\lambda = \frac{h}{mv}$$

Where

$h \rightarrow$ Planck's constant, $m \rightarrow$ mass of particle, $v \rightarrow$ velocity of the particle

7.2.2 Derivation of de-Broglie's wavelength

According to Planck's quantum theory of radiation (light) the energy of photon is given by

$$E = h\nu \longrightarrow (1)$$

Where $h \rightarrow$ Planck's constant, $\nu \rightarrow$ frequency of radiation

According to Einstein's energy-mass relation, the energy of photon is given by

$$E = mc^2 \longrightarrow (2)$$

Where $m \rightarrow$ mass of photon

C → velocity of light or photon

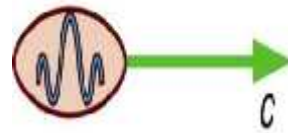


Figure.2. Motion of photon

From the above equations (1) and (2); we can write

$$h\nu = mc^2 \quad [\because C = \nu\lambda \text{ and } \nu = \frac{c}{\lambda}]$$

$$\frac{hc}{\lambda} = mc^2$$

$$\frac{h}{\lambda} = mc$$

$$\lambda = \frac{h}{mc} = \frac{h}{p} \rightarrow (3)$$

Where $p = mc$ is the momentum of photon having mass m and travelling with velocity c .

de-Broglie suggested that equation (3) can be applied both for photos and material particles. If m is mass of the particle and v is the velocity of the particle, then Momentum (p) = mv

$$\text{de-Broglie wavelength } \lambda = \frac{h}{mv} \rightarrow (4)$$

This equation represents that the de-Broglie wave length associated with a particle and also known as de-Broglie's wave equation.

7.2.3 OTHER FORMS OF de-Broglie's WAVELENGTH:

1. de-Broglie's wavelength in terms of energy

If m is mass of the particle and v is the velocity of the particle, then the kinetic energy of the particle;

$$E = \frac{1}{2}mv^2$$

Multiplying by ' m ' on both sides; we get

$$mE = \frac{1}{2}m^2v^2$$

$$2mE = m^2v^2$$

$$(\text{Or}) mv = \sqrt{2mE}$$

$$\text{de-Broglie wavelength } \lambda = \frac{h}{mv} = \frac{h}{\sqrt{2mE}} \rightarrow (5)$$

2. de-Broglie's wavelength in terms of voltage

If a charged particle of charge 'q' is accelerated through a potential difference 'V', then

$$E = qV \longrightarrow (6)$$

Also we know that the K.E of the particle $E = \frac{1}{2}mv^2 \longrightarrow (7)$

Equating equations (7) and (6) we get

$$\frac{1}{2}mv^2 = qV$$

Multiplying by 'm' on both sides we get

$$\frac{1}{2}m^2v^2 = m q V$$

$$\therefore m^2v^2 = 2m q V$$

$$mv = \sqrt{2m q V} \longrightarrow (8)$$

Substituting equation 8 in equation 4, we get

$$\therefore \text{de-Broglie wavelength } \lambda = \frac{h}{mv} = \frac{h}{\sqrt{2m q V}} \longrightarrow (9)$$

3. de-Broglie's wavelength for electron

Substituting the values of m, q and h in equation 9 we get

$$\lambda = \frac{6.626 \times 10^{-34} \text{ Js}}{\sqrt{2 \times 9.1 \times 10^{-31} \text{ Kg} \times 1.6 \times 10^{-19} \text{ C} \times V}}$$

$$= \frac{12.27 \times 10^{-10} \text{ m}}{\sqrt{V}} \quad [1 \text{ A}^0 = 10^{-10} \text{ m}]$$

$$\lambda = \frac{12.27}{\sqrt{V}} \text{ A}^0$$

This equation represents the de-Broglie wavelength associated with an electron.

4. Wave velocity

We know that

$$E = h\nu \text{ and } E = mc^2$$

$$h\nu = mc^2$$

$$v = \frac{mc^2}{h}$$

The wave velocity is given by

$$\omega = v\lambda$$

$$\omega = \frac{mc^2}{h} \cdot \frac{h}{mv}$$

$$\therefore \omega = \frac{c^2}{v}$$

Thus matter waves travel faster than velocity of light.

7.2.4 Properties of de-Broglie waves:

The following are the properties of matter waves.

1. The de-Broglie wavelength of particle of mass (m) moving with velocity (V) is given by $\lambda = \frac{h}{mv}$

2. If $v=0$, $\lambda \rightarrow \infty$ and if $v \rightarrow \infty$, $\lambda \rightarrow 0$.

This implies that matter waves are associated with material particles only if they are in motion.

3. Lesser the velocity of the particle, larger the wavelength associated with it.

4. Smaller the mass of the particle, the greater the length associated with it.

5. The matter wave of particle is independent of charge (or) nature of the particle.

6. de-

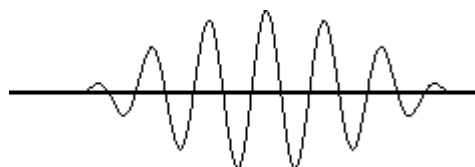
Broglie waves are pilot waves. The word, pilot means that these waves pilot (or) guide the particle.

7. Matter waves travel faster than velocity of light, i.e., $\omega = \frac{c^2}{v}$.

8. Matter waves are not electromagnetic waves.

9. The wave nature of matter introduces an uncertainty in the location of the position of the particle because a wave cannot be set exactly at this position (or) at that position.

10. de-Broglie waves are called probability waves. The amplitude of the wave tells us about the probability of finding particle in space at a particular instant. Large wave amplitude means large probability of finding the particle at that position.



7.3 Heisenberg uncertainty principle

In 1927, Heisenberg proposed a very interesting principle, which is direct consequence of the dual nature of matter, known as uncertainty principle. According to Heisenberg uncertainty principle, it is impossible to determine accurately and simultaneously the values of position and momentum of a particle at any time. If Δx is the error (uncertainty) in measuring position, Δp is the error (uncertainty) in measuring momentum; then according to this principle

$$\Delta x \cdot \Delta p \geq \frac{h}{4\pi}$$

NOTE: Similarly such relations are $\Delta E \cdot \Delta t \geq \frac{h}{4\pi}$ and $\Delta J \cdot \Delta \theta \geq \frac{h}{4\pi}$, Where ΔE and Δt are the uncertainties in determining the energy and time, while ΔJ and $\Delta \theta$ are uncertainties in determining the angular momentum and angle.

7.3.1 Application of uncertainty principle

1. It is verified Nonexistence of Electrons inside the nucleus.
2. It is also verified Existence of protons and neutrons inside the nucleus.
3. It is calculated binding energy of electron in an atom.
4. It is also calculated Mass of π -meson.
5. Emitted radiation of excited atom is confirmed by this principle.

7.4. Physical significance of wavefunction

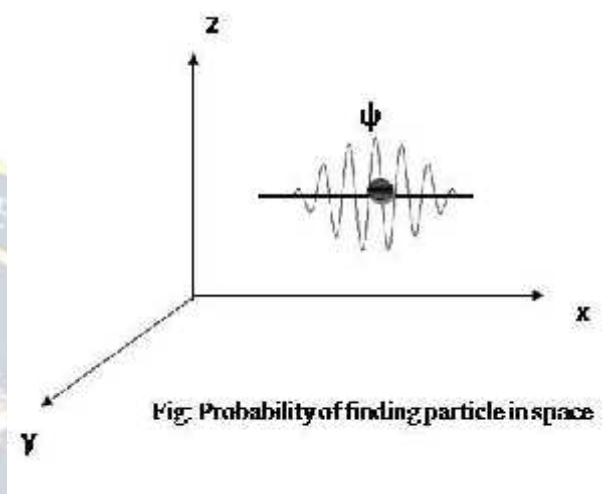
- The wave nature of matter introduced an uncertainty in the location of the Position of the particle because a wave cannot be set exactly at this position (or) at that position.
- According to Heisenberg uncertainty principle tells us that we cannot detect the particle nature and wave nature at the same time.
- To resolve this uncertainty, Eddington once upon a time humorously suggested that the primordial entity is neither particle nor wave but a 'wavicle'. Thus, de-Broglie waves are called probability waves.

- The amplitude of the wave tells us about the probability of finding particle in space at a particular instant. A large wave amplitude means a large probability of finding the particle at that position.
- Mathematically, to explain it, scientists introduced a new physical quantity called wave function.

7.4.1 Wavefunction

It is a mathematical tool used in quantum mechanics to describe any physical system, and it is denoted by ' ψ '.

It is a variable quantity i.e., associated with a moving particle at any position (x,y,z) and at any time t and it relates probability of finding the particle at that point and at that time.



- It must be well behaved, that is, single valued and continuous everywhere.
- It is usually a complex quantity and individually it has no meaning.
- To explain it, Max Born suggested a new idea about the physical significance of ψ which is generally accepted nowadays.
- According to Max Born $\psi^* \psi = |\psi|^2$ is a real and positive, it has physical meaning.
- It represents the probability of finding the particle in the state ψ .
- Since the wave function is a complex quantity, it may be expressed in the form

$$\psi(x,y,z) = (a+ib)$$

Where a, b are real functions of the variables (x,y,z,t) and $i = \sqrt{-1}$.

The complex conjugate of ψ is given by

$$\psi^*(x,y,z) = (a-ib)$$

Multiplying the above two equations, we have $\psi\psi^* =$

$$(a+ib)(a-ib) = a^2 + b^2$$

According to Max Born

$$P = \Psi \Psi^* = |\Psi_{(x,y,z)}|^2 = a^2 + b^2$$

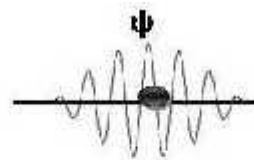


Fig. Probability of finding particle in space

Thus the product of Ψ and Ψ^* is real and positive if $\Psi \neq 0$ and is known as probability density of the particle associated with the de-Broglie wave.

The probability of finding the particle in a volume $dv = dx \cdot dy \cdot dz$ is given by

$$\text{Probability}(p) = |\Psi|^2 dx dy dz$$

For the total probability of finding the particle somewhere in space,

$$P = \iiint |\Psi|^2 dx dy dz = 1$$

A wave function Ψ satisfying the above relation is called a normalized wave function.

For motion of a particle in one dimension, the quantity

$P dx = |\Psi|^2 dx$ is the probability of finding the particle over a small distance dx at position x .

7.5 Schrodinger wave equation

Definition: The equation that describes the wave nature of a particle in mathematical form is known as Schrodinger's wave equation.

There are two types of Schrodinger wave equation, they are

- 1) Time independent wave equation and
- 2) Time dependent wave equation

7.5.1 Schrodinger one dimensional Time independent wave equation

Let us consider a particle of mass m moving with velocity v . Let ϕ be the wave function of a particle along X -direction at any time (as shown in figure).

The classical differential equation of wave motion is given by

$$\frac{\partial^2 \psi}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 \psi}{\partial t^2} \longrightarrow (1)$$

The solution for equation (1) is assumed to be

$$\psi = \psi_0 e^{-i\omega t} \longrightarrow (2)$$

Where ψ_0 is the amplitude of the wave at any

point. Differentiating equation (2) partially with respect to 't', we get

t $\longrightarrow$

$$\frac{\partial \psi}{\partial t} = \psi_0 e^{-i\omega t} \times (-i\omega) \quad (3)$$

Differentiating once again equation (3) partially with respect to 't', we get

$$\begin{aligned} \frac{\partial^2 \psi}{\partial t^2} &= \psi_0 e^{-i\omega t} \times (-i\omega) \times (-i\omega) \\ &= \psi_0 e^{-i\omega t} (-\omega^2) \quad [\text{from eq(2)}] \\ &= -\omega^2 \psi \quad \longrightarrow (4) \end{aligned}$$

Substituting equation (4) in (1) we get

$$\begin{aligned} \frac{\partial^2 \psi}{\partial x^2} &= \frac{1}{v^2} (-\omega^2 \psi) \\ &= -\omega^2 \frac{\psi}{v^2} \quad \longrightarrow (5) \end{aligned}$$

But $\omega = 2\pi\nu = 2\pi v/\lambda$ ($v = \nu\lambda$ and $\nu = v/\lambda$)

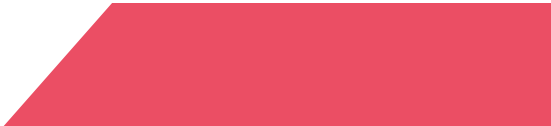
$$(\text{or}) \frac{\omega}{v} = \frac{2\pi}{\lambda} \longrightarrow (6)$$

Substituting equation (6) in (5), we get

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{4\pi^2 \psi}{\lambda^2} = 0 \longrightarrow (7)$$

This is the classical wave equation in terms of wavelength.

Schrodinger connected the expression for the de Broglie wave the length in to the classical wave equation in terms of wavelength for a moving particle and obtained a new



We know that de Broglie wavelength is given by

$$\lambda = \frac{h}{mv} \rightarrow (8)$$

Substituting equation (8) in (7), we get

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{4\pi^2 m^2 v^2}{h^2} \psi = 0 \rightarrow (9)$$

If E is the total energy of a particle, ' V ' is the potential energy and $\frac{1}{2}mv^2$ is the kinetic energy; then

$$\text{Total energy (E)} = P.E + K.E$$

$$E = V + \frac{1}{2}mv^2$$

$$\text{(Or)} \quad E - \frac{1}{2}mv^2 = 0$$

$$2(E - V) = mv^2$$

Multiplying by ' m ' on both sides in the above equation, we get

$$2m(E - V) = m^2 v^2 \rightarrow (10)$$

Substituting equation (10) in (9) we get

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{8\pi^2 m}{h^2} (E - V) \psi = 0 \rightarrow (11)$$

$$\text{Or} \quad \frac{\partial^2 \psi}{\partial x^2} + \frac{2m}{\hbar^2} (E - V) \psi = 0 \rightarrow (12)$$

Where $\hbar = \frac{h}{2\pi}$ is called Planck's reduced constant.

This equation is known as Schrodinger time independent one-dimensional wave equation.

Note:

For 3 dimensional:

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \frac{\partial^2 \psi}{\partial z^2} + \frac{8\pi^2 m}{h^2} (E - V) \psi = 0$$

$$\text{Or } \nabla^2 \psi + \frac{8\pi^2 m}{h^2} (E - V) \psi = 0$$

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \text{ is called Laplace operator.}$$

7.5.2 Schrodinger one dimensional Time independent wave equation

Let us consider a particle of mass m moving with velocity v . Let ψ be the wavefunction of a particle along x -direction at any time (as shown in figure)

The classical differential equation of wave motion is given by

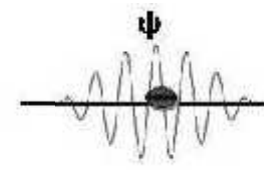


Fig: Probability of finding particle in space

$$\frac{\partial^2 \psi}{\partial x^2} = -\frac{1}{v^2} \frac{\partial^2 \psi}{\partial t^2} \quad \longrightarrow \quad (1)$$

The solution for equation (1) is assumed to be

$$\psi = \psi_0 e^{-i\omega t} \quad \longrightarrow \quad (2)$$

Where ψ_0 is the amplitude of the wave at any point. Differentiating equation (2) partially with respect to 't', we get

$$\begin{aligned} \frac{\partial \psi}{\partial t} &= \psi_0 e^{-i\omega t} \times (-i\omega) \\ &= -i\omega \psi \quad \longrightarrow \quad (3) \end{aligned}$$

$$\begin{aligned} \text{But } E &= h\nu = h\omega \frac{1}{2\pi} \\ &= \frac{h}{2\pi} \omega \quad (\omega = 2\pi\nu \text{ and } \nu = \frac{\omega}{2\pi}) \end{aligned}$$

$$\therefore E = \hbar\omega$$

$$\text{And } \omega = \frac{E}{\hbar} \quad \longrightarrow \quad (4)$$

Substituting equation (4) in eq (3), we get

$$\begin{aligned} \frac{\partial \psi}{\partial t} &= -i \frac{E}{\hbar} \psi \\ &= -i \frac{E}{\hbar} \psi \\ E\psi &= i\hbar \frac{\partial \psi}{\partial t} \quad \longrightarrow \quad (5) \end{aligned}$$

Substituting equation (5) in Schrodinger time independent one-

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{2m}{\hbar^2} (E - V) \psi = 0 \longrightarrow (6)$$

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{2m}{\hbar^2} (E \psi - V \psi) = 0$$

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{2m}{\hbar^2} \left(i\hbar \frac{\partial \psi}{\partial t} - V \psi \right) = 0$$

Multiplying $\frac{\hbar^2}{2m}$ on both sides, we get

$$\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + \frac{\hbar^2}{2m} \left(i\hbar \frac{\partial \psi}{\partial t} - V \psi \right) = 0$$

$$\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + \left(i\hbar \frac{\partial \psi}{\partial t} - V \psi \right) = 0$$

$$\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} - V \psi = -i\hbar \frac{\partial \psi}{\partial t}$$

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + V \psi = i\hbar \frac{\partial \psi}{\partial t}$$

This equation is known as Schrodinger time dependent one-dimensional wave equation.

$$\text{Or} \quad H\psi = E\psi$$

Where $H \equiv -\frac{\hbar^2 \partial^2}{2m \partial x^2} + V$ is called Hamilton operator

$E = i\hbar \frac{\partial}{\partial t}$ is called energy operator.

7.6 Particle in one dimensional box or well

- Let us consider a particle of mass 'm' moving with velocity 'v' along x-direction and is confined between two infinite potential rigid walls, so that the particle has no chance of escaping from them. Therefore, the particle bounces back and forth between two walls as shown in fig.

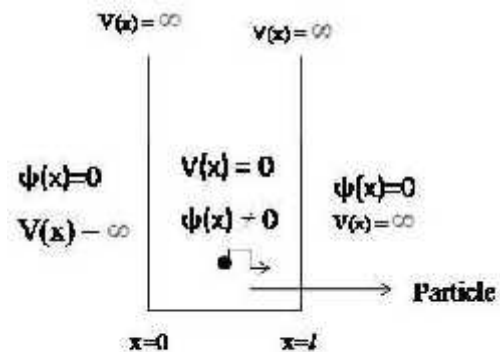


Fig: Particle in one dimensional box

- Let the potential energy of electron inside the box is constant and can be taken as zero for simplicity.
- From fig, we consider two regions:

(i) Outside the box

(ii) Inside the box

Outside the box:

The particle cannot exist outside and on the walls of the box; therefore the probability of finding the electron outside the wall is zero and the potential energy is ∞ .

i.e.,

$$\begin{matrix} \psi(x)=0 \\ V(x)=\infty \end{matrix}$$

When $x \leq 0$ and $x \geq l$

Inside the box:

The particle exists inside the box; therefore the probability of finding the electron inside the box is not equal to zero and the potential energy is zero.

i.e.,

$$\begin{matrix} \psi(x) \neq 0 \\ V(x) = 0 \end{matrix}$$

When $x > 0$ and $x < l$

- To calculate the probability of finding particle within the box, let us consider one dimensional time independent Schrödinger wave equation

$$\text{i.e., } \frac{d^2\psi}{dx^2} + \frac{8\pi^2m}{h^2}(E - V(x))\psi = 0 \longrightarrow (1)$$

Inside the box, the potential energy $V(x) = 0$

Therefore, equation (1) becomes

$$\frac{d^2\psi}{dx^2} + \frac{8\pi^2mE}{h^2} \psi = 0$$

$$\text{(Or) } \frac{d^2\psi}{dx^2} + k^2\psi = 0 \longrightarrow (2)$$

$$\text{Where } k^2 = \frac{8\pi^2mE}{h^2} \longrightarrow (3)$$

Equation (2) is a second order differential equation; therefore it should have solution with two arbitrary constants.

∴ The solution for equation (2) is given by $\Psi(x) = A$

$$\sin Kx + B \cos Kx \longrightarrow (4)$$

Where A and B are called as arbitrary constants, which can be found by applying the boundary conditions.

Boundary Condition at $x=0$:

At $x=0$, the probability of finding the electron is zero, i.e., $\psi(x)=0$. Equation (4) becomes

$$0 = A \sin K(0) + B \cos K(0) \quad 0 = 0 + B$$

$$\therefore B = 0 \longrightarrow (5)$$

Boundary Condition at $x=l$:

At $x=l$, the probability of finding the electron is zero, i.e., $\psi(x)=0$. Equation (4) becomes

$$0 = A \sin K(l) + B \cos K(l)$$

$$0 = A \sin K(l) + (0) \cos K(l)$$

$$0 = A \sin K(l)$$

$$\text{Since } A \neq 0; \sin K(l) = 0$$

[Since $B = 0$]

(6)

$$\text{We know that, } \sin n\pi = 0$$

(7)

Comparing these two equations, we can write $K(l) =$

$$n\pi$$

$$K = \frac{n\pi}{l} \longrightarrow (8)$$

$$k^2 = \frac{n^2 \pi^2}{l^2} \longrightarrow (9)$$

Substituting the values of ' B ' and ' K ' in equation (4),

Equation (4) becomes

$$\Psi(x) = A \sin \frac{n\pi x}{l} \longrightarrow (10)$$

This equation represents the wavefunction associated with moving free electron inside the box.

Normalization of the wave function: it is the process by which the probability (P) of finding the particle or electron inside the box can be done.

The total probability that the particle is somewhere in the box must be unity

$$\text{i.e., } P = \int_0^l |\Psi_n(x)|^2 dx \longrightarrow (11)$$

[Since the particle is present inside the box between the length 0 to l, then the limits are chosen between 0 to l]

Substituting equation (10) in equation (11), we get

$$\begin{aligned} P &= \int_0^l A^2 \sin^2 \frac{n\pi x}{l} dx = 1 \\ A^2 \int_0^l \sin^2 \frac{n\pi x}{l} dx &= 1 \quad \left[\text{Since } \sin^2 x = \frac{1 - \cos 2x}{2} \right] \\ A^2 \int_0^l \frac{1 - \cos 2n\pi x}{2} dx &= 1 \quad \left[\text{Since } \int_0^l \cos n\pi x dx = \frac{\sin n\pi x}{n\pi} \right] \\ \frac{A^2}{2} \int_0^l \frac{1 - \cos 2n\pi x}{1} dx &= 1 \\ \frac{A^2}{2} \int_0^l \frac{1 - \cos 2n\pi x}{1} dx &= 1 \\ \frac{A^2}{2} \left[\int_0^l 1 dx - \int_0^l \frac{\cos 2n\pi x}{1} dx \right] &= 1 \\ \frac{A^2}{2} [1] - \frac{A^2}{2} \left[\frac{\sin 2n\pi l}{2n\pi} - \frac{\sin 2n\pi \cdot 0}{2n\pi} \right] &= 1 \quad (12) \end{aligned}$$

Since $\sin n\pi = 0$, $\sin 2n\pi$ is also =

0 Equation (13) can be written as

$$\frac{A^2}{2} [1] = 1$$

$$A^2 = \frac{2}{1}$$

$$\boxed{A = \sqrt{\frac{2}{l}}} \quad (13)$$

Substituting the values of 'A' in equation (10),

$$\Psi_n(x) = \sqrt{\frac{2}{l}} \sin \frac{n\pi x}{l} \quad (14)$$

Each value of wave function is known as Eigenfunction. The various Eigenfunctions of a free electron enclosed in a 1D box as shown in fig.

Case 1: If $n=1$, then $\Psi_1(x)$ has two nodes at $x=0$

$x=l$.

Case 2: If $n=2$, then $\Psi_2(x)$ has two nodes at $x=0$

$x=\frac{l}{2}$ and $x=l$.

Case 3: If $n=3$, then $\Psi_3(x)$ has three nodes at

$x=0$, $x=\frac{l}{3}$, $x=\frac{2l}{3}$ and $x=l$. Therefore $\Psi_n(x)$ has $(n+1)$ nodes.

Probability of the location of the particle:

The probability of finding a particle over a small distance dx at x is given by

$$P(x)dx = |\Psi_n(x)|^2 dx$$

$$P(x)dx = \frac{2}{l} \sin^2 \frac{n\pi x}{l} dx$$

Thus, the probability density for one-dimensional motion is

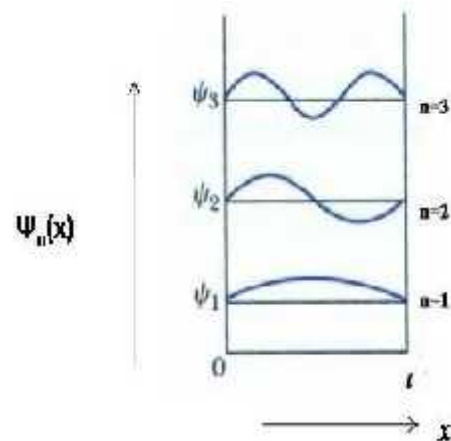
$$P(x) = \frac{2}{l} \sin^2 \frac{n\pi x}{l}$$

The probability density is maximum when $\frac{n\pi x}{l} = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \dots$

$$x = \frac{l}{2n}, \frac{3l}{2n}, \frac{5l}{2n}, \dots$$

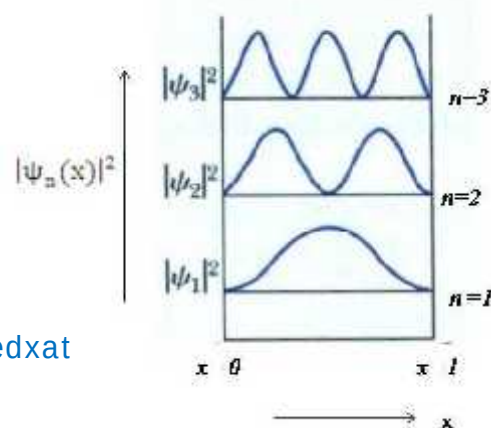
Case 1: If $n=1$, then the probability of position of the particle is at $x=\frac{l}{2}$

Case 2: If $n=2$, then the probability of position of the particle is at $x=\frac{l}{4}$ and $x=\frac{3l}{4}$.



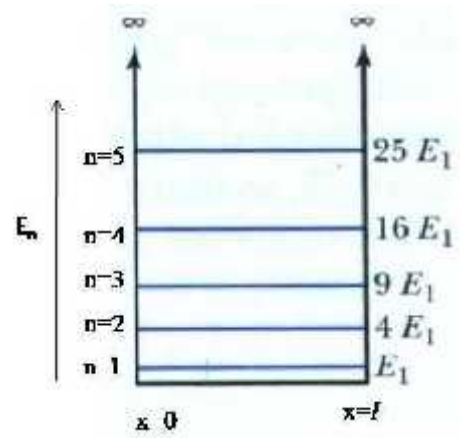
and

and



$x=0$

Case3: If $n=3$, then the probability of position of the particle is at $x = \frac{l}{6}$ and $\frac{5l}{6}$. these positions are shown in fig.



Energy of the Particle:

From the equations (3) & (9)

$$k^2 = \frac{8\pi^2 m E}{h^2} \rightarrow (3)$$

$$k^2 = \frac{n^2 \pi^2}{l^2} \rightarrow (9)$$

$$\frac{8\pi^2 m E}{h^2} = \frac{n^2 \pi^2}{l^2}$$

$$E_n = \frac{n^2 h^2}{8ml^2} \rightarrow (15)$$



This is an expression for the energy of the particle. Each value of E_n is known as Eigenvalue.

The various Eigenvalues of an electron enclosed in a 1D box are shown in fig

Case1: If $n=1$, then $E_1 = \frac{h^2}{8ml^2}$

Case2: If $n=2$, then $E_2 = \frac{4h^2}{8ml^2} = 4E_1$

Case3: If $n=3$, then $E_3 = \frac{9h^2}{8ml^2} = 9E_1$ and soon. Therefore, energy levels of electron are discrete.

7.7 Introduction to free electron theory

The study of behaviour of free electrons and various properties such as electrical, thermal and magnetic properties etc., in a solid is known as free electron theory of solids.

The electron theory of solids has been developed in three main stages.

(i) The classical free electron theory

- This theory was developed by Drude and Lorentz in 1900.
- According to this theory, metals consist of positive ion cores and free electrons.
- The free electrons bound to move within the metal in a completely uniform potential field among these ion cores and obey the laws of classical mechanics.

(ii) The Quantum free electron theory

- This theory was developed by Sommerfeld in 1928.
- The free electrons bound to move within the metal in a completely uniform potential field among the ion cores and obey quantum laws.

(iii) The Zone theory

- Bloch stated this theory in 1928.

According to this theory, the free electrons move in a periodic field provided by the lattice and obey quantum laws. This theory is also called "Band theory of solids".

7.8 Merits of classical free electron theory

- It verifies Ohm's law.
- It explains the electrical conductivity of metals.
- It explains the electrical resistivity of metals.
- It explains the thermal conductivities of metals.
- It derives Weidman's-Franz law. (i.e., the relation between electrical conductivity and thermal conductivity)
- It explains optical properties of metals.

7.9 Demerits of classical free electron theory

- This theory fails to explain the photoelectric effect, Compton Effect and the black body radiation.

- This theory fails to explain specific heat of metals.
- According to this theory, the electrical conductivity (σ) of a metal is inversely proportional to the square root of its absolute temperature (T), i.e., $\sigma \propto \frac{1}{\sqrt{T}}$.

But practically, it has been observed that the electrical conductivity is inversely proportional to its absolute temperature, i.e., $\sigma \propto \frac{1}{T}$.

- Thus, this theory failed to explain electrical conductivity of metals.
- This theory fails to explain mean free path of electrons. Because the theoretical and experimental values of mean free path (λ) do not coincide.

7.10 Quantum free electron theory

- This theory was developed by Sommerfeld in 1928 based on the Pauli Exclusion Principle and Fermi-Dirac Statistics.
- According to this theory, metals consist of positive ion cores and free electrons.
- The free electrons bound to move within the metal in a completely uniform potential field among these ion cores and obey quantum laws.
- Mutual attraction between electrons and positive ions and the repulsion between individual electrons may be ignored.
- The free electrons will have discrete energy states (or) levels in a metal.
- The occupation (or) distribution of electrons among these energy levels will be governed by Pauli's exclusion principle.
- According to Pauli's exclusion principles, not two electrons can have the same set of quantum numbers in a given energy state, i.e., not two electrons can occupy the same energy level and each energy can accept only two electrons, one with spin up ($\uparrow$) and the other with spin down ($\downarrow$).
- At absolute zero temperature (0K), the first electron can occupy the lowest available energy state (E_0) with spin up ($\uparrow$) and the next electron can also occupy the same energy level with opposite spin down ($\downarrow$) of first electron. The third electron cannot occupy in this energy level, because of Pauli's exclusion principle and it will go to the next energy (E_1). In this way, all free electrons can be

occupied in the different energy states in the

dimensional potential box as shown in figure.

- The highest energy level occupied by electrons at absolute zero temperature is known as "Fermi energy level" (E_F). Above Fermi energy level all energy levels are empty. i.e., at 0K all the energy levels below Fermi energy level are

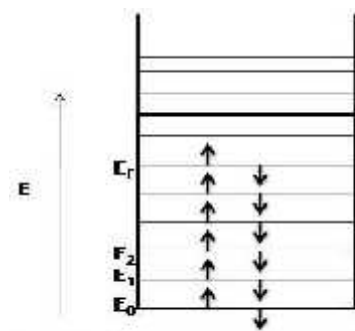


Fig: Distribution of free electrons in various energy levels at 0K.

one

filled

by free electrons. So they possess different energies and velocities or momentums.

- The different velocities of these free electrons of a metal can be represented in velocity space as shown in fig.
- At 0K, free electrons present in Fermi energy level possess maximum velocity represented by V_F .
- We assume a sphere of radius V_F at the origin of velocity space as shown in fig.
- At 0K, all points inside the Fermi sphere are occupied.
- Each point inside the sphere from the origin represents the velocity of a free electron.
- The vectors joining different points inside the sphere from the origin represent velocity vectors.
- In the absence of electric field, the velocity vectors in the Fermi sphere present from origin will be cancelled with each other and the net velocity of electrons in all directions is zero.
- If the electric field E is applied on the free electrons along positive X -direction, then the electrons experience a Lorentz force F_L along negative X -direction.
- The electrons which are present near the Fermi surface take electrical energy and occupy higher vacant energy levels as a result the sphere slightly displaces as shown in fig. The remaining electrons are unable to occupy higher vacant energy levels because the Lorentz force is less on these electrons.

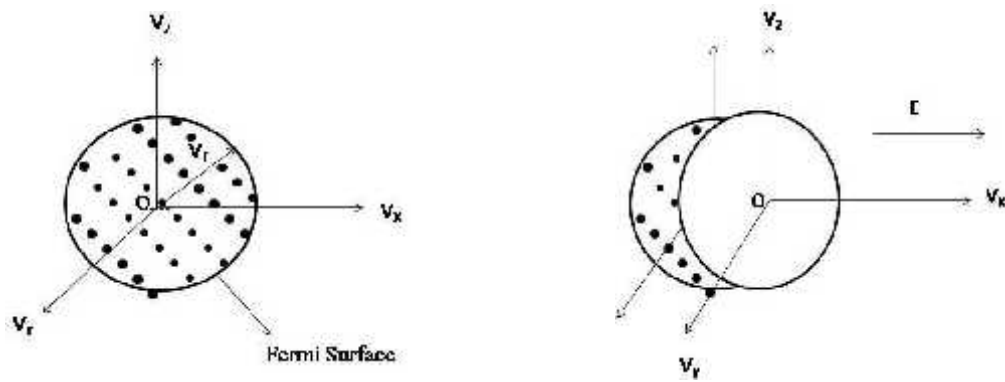


Fig: Displacement of Fermi sphere under applied electric field.

- Under an electric field:

Each electron experiences a Lorentz force $F_L = eE$

(1)

From Newton's Second law of motion,

The force on the electron $F = ma$

(2)

Equating equation (2) and equation (1), we have

$$ma = eE \quad \longrightarrow \quad (3)$$

The relation between momentum p and the wave or propagation vector k is given by

$$p = \hbar k \quad \longrightarrow \quad (4)$$

$$\text{But } p = mv \quad \longrightarrow$$

(5) Equ

ating equation (4) and equation (5), we have

$$mv = \hbar k$$

$$v = \frac{\hbar k}{m} \quad \longrightarrow \quad (6)$$

Differentiating equation (6) with respect to time gives acceleration,

$$a = \frac{dv}{dt} = \frac{\hbar}{m} \frac{dk}{dt} \quad \longrightarrow \quad (7)$$

Substituting equation (7) in (3), we get

$$m_{\text{mdt}}^{\text{hdk}} = eE \longrightarrow (8)$$

$$dk = \frac{eE}{\hbar} dt \longrightarrow (9)$$

Integrating the equation (9), taking limits from 0 to t;

$$\int_0^t dk = \int_0^t \frac{eE}{\hbar} dt$$

$$K(t) - K(0) = \frac{eEt}{\hbar} \longrightarrow (10)$$

Let the mean collision time and mean free path of electron present at Fermi surface is represented as τ_F and λ_F respectively then, we have:

$$\tau_F = \lambda_F / v_F \longrightarrow (11)$$

For an electron at Fermi surface, $t = \tau_F$ and $K(t) - K(0) = \Delta k$.

$$\text{Then } \Delta k = \frac{eE\tau_F}{\hbar} \longrightarrow (12)$$

Substituting equation (11) in (12), we get

$$\Delta k = \frac{eE}{\hbar} \left[\frac{\lambda_F}{v_F} \right] \longrightarrow (13)$$

If the increased velocity is Δv due to application of electric field then the current density $J = ne\Delta v$

$$\longrightarrow (14)$$

Where n is the number of free electrons per unit volume and e is the charge of the electron.

From equations (6) and (13), analytically Δv can be expressed as

$$\Delta v = \frac{\hbar \Delta k}{m^*} = \frac{eE}{m^* \hbar} \left[\frac{\lambda_F}{v_F} \right] = \frac{eE}{m^*} \left[\frac{\lambda_F}{v_F} \right] \longrightarrow (15)$$

Where m^* is the effective mass of free electron. Substituting equation (15) in (14), we get

$$J = ne \times \frac{eE}{m^*} \left[\frac{\lambda_F}{v_F} \right]$$

$$= ne \frac{e^2 E}{m^*} \left[\frac{\lambda_F}{v_F} \right] \longrightarrow (16)$$

$$\text{The electrical conductivity } \sigma = \frac{J}{E} \longrightarrow (17)$$

Substituting equation (16) in (17), we get

$$\sigma = \frac{ne^2 E}{m^*} \left[\frac{\lambda_F}{v_F} \right]$$

$$= \frac{ne^2}{m^*} \left[\frac{\lambda_F}{v_F} \right] \quad [\text{From eq(11)}]$$



This is the expression for electrical conductivity which is found to be in good agreement with experimental values.

Thus we can say that, according to this theory, all free electrons are not responsible for electrical conduction, only those free electrons which are very near to Fermi surface.

7.10.1 Merits of Quantum free electron theory

- It explains the electrical conductivity of metals.
- It explains the electrical resistivity of metals.
- It explains the thermal conductivities of metals.
- It derives Wiedemanns-Franz law. (i.e., the relation between electrical conductivity and thermal conductivity)
- It explains optical properties of metals.
- It successfully explains the photo electric effect, Compton Effect and the black body radiation etc.
- It explains the specific heat of metals.

7.10.2 Demerits of Quantum free electron theory

- This theory fails to explain the differences between insulators, conductors and semiconductors.
- Positive hall coefficient of metals and
- Lower conductivities of divalent metal than monovalent metals.

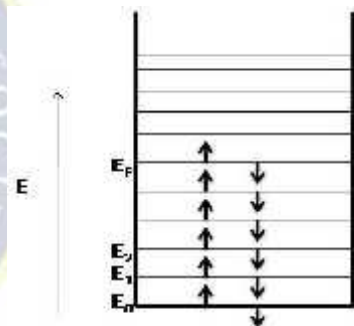


Fig. Distribution of free electrons in various energy levels at 0K.

7.11 Fermi-Dirac distribution:

- According to Fermi-Dirac statistics, electrons are indistinguishable and half integral spin particles which obey the Pauli's exclusion principle. They are known as fermions (or) Fermi particles.
- The Fermi-Dirac distribution describes the behaviour of free electrons in a metal, based on the quantum mechanics and Pauli's exclusion principle.
- According to quantum mechanics, the electrons will have discrete energy states (or) level and the occupation (or) distribution of electrons among these energy levels will be governed by Pauli's exclusion principle.

- According to Pauli's exclusion principles, not two electrons can have same set of quantum numbers in a given energy state. i.e., each energy can accept only two electrons one with spin up ($\uparrow$) and other with spin down ($\downarrow$).
- The occupation of free electrons in one-dimensional potential well as shown in fig. At absolute zero temperature (0K), the first electron can occupy the lowest available energy state (E_0) with spin up ($\uparrow$) and then next electron can also occupy the same energy level with opposite spin down ($\downarrow$) of first electron. The third electron cannot occupy in this energy level, because of Pauli's exclusion principle and it will go to the next energy (E_1). In this way, all free electrons can be occupied in the different energy states in the one-dimensional potential box as shown in figure.

- Fermi Energy Level (E_F): -

The highest energy level occupied by electrons at absolute zero temperature is known as "Fermi energy level" (E_F) which divides (or) separates the occupied states from the unoccupied states.

Fermi-Dirac distribution function:-

The Fermi-Dirac distribution function gives the probability of occupation of energy levels by the electrons (fermions).

- The probability $F(E)$ of occupation of particular energy level (E) by the electrons at temperature (T) is given by

$$F(E) = \frac{1}{1 + \exp\left(\frac{E - E_F}{k_B T}\right)}$$

Where E = Particular energy level (eV) E_F

= Fermi energy level (eV)

k_B = Boltzmann constant = 1.38×10^{-23} J/K. T = Temperature (K).

7.11.1 Effect of Temperature on F -

Distribution Function Case 1: Probability of occupation at $T = 0$

K, When $E < E_F$

$E < E_F$ means energy levels lying below E_F , the term $\frac{E - E_F}{k_B T}$

takes a negative value. Hence equation (1) becomes

$$\frac{E - E_F}{k_B T} = -\infty \text{ at } T = 0 \text{ K, When } E < E_F$$

$$F(E) = \frac{1}{1 + \exp^{-\infty}} = \frac{1}{1 + 0} = 1 \quad [\text{Since } \exp^{-\infty} = 0]$$

$$\therefore F(E)=1$$

It implies that, all the energy levels below E_F are occupied by free electrons.

Case 2: Probability of occupation at $T = 0\text{K}$, When $E > E_F$

$E > E_F$ means energy levels lying above E_F , the term $\frac{E - E_F}{k_B T}$ takes a positive value. Hence equation (1) becomes

$$\frac{(E - E_F)}{k_B T} = \infty \text{ at } T = 0\text{K, When } E > E_F.$$

$$F(E) = \frac{1}{1 + \exp^{\infty}} = \frac{1}{1 + \infty} = \frac{1}{\infty} = 0 \text{ [Since } \exp^{\infty} = \infty \text{]}$$

$$\therefore F(E) = 0$$

It implies that, all the energy levels above E_F are unoccupied by free electrons.

Case 3: Probability of occupation at $T > 0\text{K}$, When $E = E_F$

The term $\frac{(E - E_F)}{k_B T} = 0$ at $T > 0\text{K}$, When $E = E_F$. Hence equation (1) becomes

$$F(E) = \frac{1}{1 + \exp^0} = \frac{1}{1 + 1} = 0.5 \text{ [Since } \exp^0 = 1 \text{]}$$

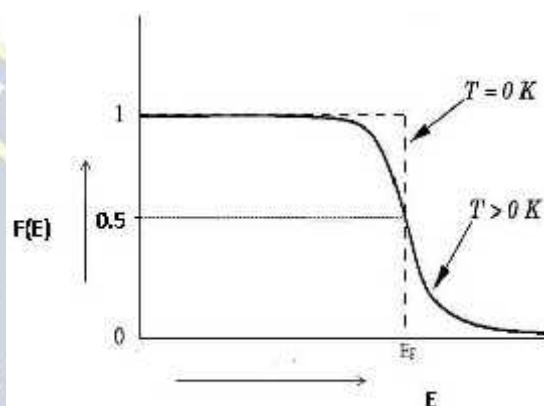
$$\therefore F(E) = 0.5$$

It implies that, the probability of occupancy of 50% at any temperature above 0K.

Graph: A graph is plotted between $F(E)$ and E at $T = 0\text{K}$ and $T > 0\text{K}$ as shown in fig. From the curve, we can observe that the curve is a step-like character at $T = 0\text{K}$. When $E > E_F$ and $E < E_F$.

This implies that all the energy levels below E_F are filled and above E_F are empty.

Further, for $T > 0\text{K}$, the Fermi function $F(E)$ smoothly falls to zero as shown in fig.



8. Practice Quiz

- Dual nature of matter was proposed by [a]
a deBroglie b Planck c Einstein d Newton
- The wavelength of de Broglie wave associated with a moving particle is independent of its [b]
a momentum b charge c velocity d mass
- The wavelength of electron moving with velocity of 500m/s is [a]
a 1.45nm b 0.50nm c 2.90nm d 3.00nm
- The value of Fermi function in Fermi level is [a]
a 0.5 b 0 c 1 d 0.75
- If the valence band is partially filled then the solid is a/an [a]
a conductor b Insulator c Both a & b d none

6. In an allowed band, the velocity of a free electron is zero at _____ [c]
 a bottom b Top c Bottom & top d None
7. The walls of a particle in a box are supposed to be _____ [d]
 a Small but infinitely hard b Infinitely large but soft c Soft and Small d Infinitely hard and infinitely large
8. The wave function of the particle lies in which region? [c]
 a $x > 0$ b $x < 0$ c $0 < x < L$ d $x > L$
9. The particle loses energy when it collides with the wall. [b]
 a true b false
10. The Energy of the particle is proportional to _____ [c]
 a n b n^{-1} c n^2 d n^{-2}
11. For a particle inside a box, the potential is maximum at $x =$ _____ [a]
 a L b $2L$ c $L/2$ d $3L$
12. The Eigen value of a particle in a box is _____ [d]
 a $L/2$ b $2/L$ c $\sqrt{2/L}$ d $\sqrt{2}/L$



9. Assignment Questions:

S.No	Question	BL	CO
1	Explain de Broglie hypothesis of matter waves. Mention its properties.	2	4
2	Explain Heisenberg's uncertainty principle along with its applications.	2	4
3	Derive the Schrodinger time dependent and independent wave equations?	6	4
4	Derive the Energy of freely moving electron in one dimensional potential box.	6	4

5	ExplainFermi-Diracdistributionfunctionalongwithitstemperaturedependence.	2	4
6	Explaintheoriginofenergybandsinsolids.	2	4

10.PartA-Question&Answers

S.No	Question&Answers	BL	CO
1	Defineclassicalmechanics. Answer: Classical mechanics deals with the motion and behaviour ofmacroscopicparticles.Ex:Motionofplanets.	1	4
2	Definequantummechanics. Answer:Quantummechanicsdealswiththemotionandbehaviourofmicroscopicparticles.Ex:MotionofElectron.	1	4
3	Summarizede-Broglie matterwaves. Answer:Thewaveassociatedwithmovingparticleofmatter(likeelectrons,potonsetc)areknownasmatterwaveorpilotwavesordeBrogliewaves.	2	4
4	Formulatede-Brogliewavelengthforelectron. Answer:Substitutingthevaluesofm,qandhinequation9weget $\lambda = \frac{6.626 \times 10^{-34} \text{ Js}}{\sqrt{2 \times 9.1 \times 10^{-31} \text{ Kg} \times 1.6 \times 10^{-19} \text{ CV}}}$ $= \frac{12.27 \times 10^{-10} \text{ m}}{\sqrt{V}} \quad [1 \text{ A}^0 = 10^{-10} \text{ m}]$ $\lambda = \frac{12.27}{\sqrt{V}} \text{ A}^0$	6	4
5	Listtheapplicationofmatterwaves. Answer.Theelectronmicroscopeisanapplicationofmatterwaves,associatedwiththeelectrons	1	4

6	Discuss the properties of de-Broglie waves? Answer: Some properties of de-Broglie waves are a) The de-Broglie wavelength of a particle of mass (m) moving with velocity (V) is given by $\lambda = \frac{h}{mv}$ If $v=0$, $\lambda \rightarrow \infty$ and if $v \rightarrow \infty$, $\lambda \rightarrow 0$. This implies that matter waves are associated with material particles only if they are in motion b) Lesser the velocity of the particle, larger the wavelength associated with it. c) Smaller the mass of the particle, the greater the length associated with it. d) The matter wave of a particle is independent of charge (or) nature of the particle. e) De-Broglie waves are pilot waves. The word, pilot means that these waves pilot (or) guide the particle. f) Matter waves travel faster than velocity of light, i.e., $v = \frac{c}{\sqrt{1 - \frac{v^2}{c^2}}}$. g) Matter waves are not electromagnetic waves. h) The wave nature of matter introduces an uncertainty in the location of the position of the particle because a wave cannot be set exactly at this position (or) at that position.	2	4
7	Explain the Heisenberg uncertainty principle. Answer: According to Heisenberg uncertainty principle, it is impossible to determine accurately and simultaneously the values of position and momentum of a particle at any time. If Δx is the error (uncertainty) in measuring position, Δp is the error (uncertainty) in measuring momentum; then according to this principle $\Delta x \cdot \Delta p \geq \frac{h}{4\pi}$	2	4
8	Examine the significance of wave function? Answer: It is a variable quantity i.e., associated with a moving particle at any position (x, y, z) and at any time t and it relates the probability of finding the particle at that point and at that time. It is usually a complex quantity and individually it has no meaning. According to Max Born $\psi \cdot \psi^* = \psi ^2$ is a real and positive quantity, it has physical meaning. It represents the probability of finding the particle in the state ψ.	3	4
9	Describe the postulates of classical free electron theory. Answer: According to this theory	2	4

	a) This theory was developed by Drude and Lorentz in 1900. b) According to this theory, metals consist of positive ion cores and free electrons. c) The free electrons bound to move within the metal in a completely uniform potential field among these ion cores and obey the laws of classical mechanics.		
10	Explain the postulates of band theory. Answer: According to this theory a) Bloch stated this theory in 1928. b) According to this theory, the free electrons move in a periodic field provided by the lattice and obey quantum laws. This theory is also called "Band theory of solids".	2	4
11	List the merits of classical free electron theory? Answer: the merits of classical free electron theory are a) It verifies Ohm's law. b) It explains the electrical conductivity of metals. c) It explains the electrical resistivity of metals. d) It explains the thermal conductivities of metals. e) It derives Weidman's-Franz law. (i.e., the relation between electrical conductivity and thermal conductivity) f) It explains optical properties of metals.	1	4
12	Derive the Fermi-Dirac distribution function. Answer: The probability $F(E)$ of occupation of a particular energy level (E) by the electrons at temperature (T) is given by $F(E) = \frac{1}{1 + \exp\left(\frac{E - E_F}{k_B T}\right)}$ Where E = Particular energy level (eV) E_F = Fermi energy level (eV)	6	4

$K_B = \text{Boltzmann constant} = 1.38 \times 10^{-23} \text{ J/K}$.

$T = \text{Temperature (K)}$.

11. Part B-Question

S.No	Question	BL	CO
1	Explain the de-Broglie hypothesis. List the properties of matter waves.	2	4
2	Calculate the wavelength associated with an electron raised to a potential of 1600V.	3	4
3	Derive Schrödinger 1D time independent wave equation.	6	4
4	Explain the physical significance of wave function.	2	4
5	Derive Energy of particle in a one-dimensional box.	6	4
6	An electron is trapped in a one-dimensional box of length $1 \times 10^{-10} \text{ m}$. Find the energy values in the ground and the first two excited states.	3	4
7	Explain the Fermi-Dirac distribution function varies with temperature.	2	4
8	Calculate the temperature at which there is 1% probability that a state with the energy 0.5 eV is above Fermi energy.	3	4

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12. Prescribed Text Books & Reference Books

Text Book

1. A text book of Engineering Physics –
Dr. M. N. Avadhanulu & Dr. P. G. Kshirsagar, S. Chand and Company, 11 Edition, 2019
2. Engineering Physics –

B. K. Pandey and S. Chaturvedi, Cengage Learning, 2013 References

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Shatendra Sharma, Jyotsna Sharma, Pearson Education, 2018 Engineering Physics –
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